

SECURITY ENGINEERING — HANDS-ON LAB

Automated Sign-In Risk Detection & Response

Microsoft Entra ID → Microsoft Sentinel → Logic Apps

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github.com/davidrufinobotte/entra-sentinel-detection-lab

HOW IT WORKS

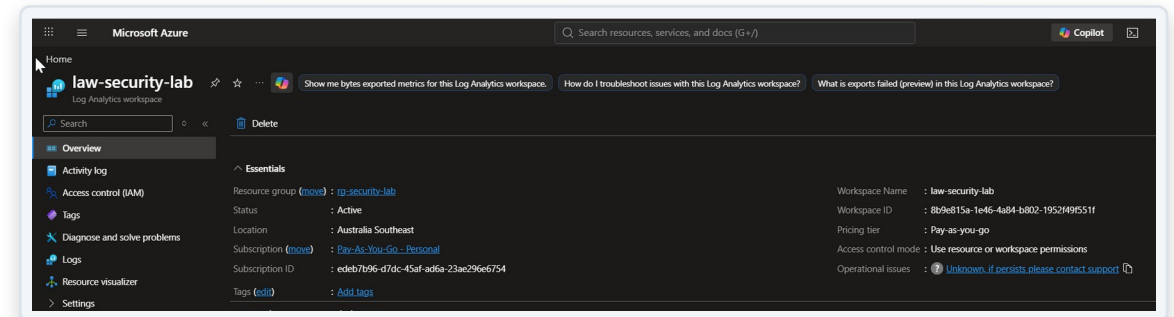
End-to-end pipeline: log to alert



A risky sign-in is logged, ingested into Sentinel, evaluated by a scheduled detection rule, and — if it matches — automatically opens an incident and triggers an emailed alert.

Log Analytics Workspace

- Created law-security-lab as the central log repository for the lab
- Pay-As-You-Go subscription, Australia Southeast region
- Base layer every downstream Sentinel component reads from



Connecting Identity Signals

- Connected the Microsoft Entra ID data connector (Sign-In + Audit logs)
- Fixed a real issue: the connector was missing until the Entra ID solution was installed from the Sentinel Content Hub
- Confirms troubleshooting ability, not just following steps

The screenshot displays the Microsoft Defender console interface. The left sidebar shows the navigation menu with 'Data connectors' highlighted. The main content area is titled 'Connector details' for the 'Microsoft Entra ID' connector. The 'Status' section shows the connector is 'Connected' and '3 Hours Ago Last Log Received'. The 'Prerequisites' section lists requirements for integration. The 'Table management' section shows a table of connectors with columns for Table name, Tier, Table type, Analytics retention, Total retention, and Workspace. The 'Configuration' section shows the 'Sign-In Logs' and 'Audit Logs' options checked.

Connector details

Microsoft Entra ID

Connected Status: Microsoft Provider, 3 Hours Ago Last Log Received

Description: Gain insights into Microsoft Entra ID by connecting Audit and Sign-in logs to Microsoft Sentinel to gather insights around Microsoft Entra ID scenarios. You can learn about app usage, conditional access policies, legacy auth relate details using our Sign-in logs. You can get information on your Self Service Password Reset (SSPR) usage, Microsoft Entra ID Management activities like user, group, role, app management using our Audit logs table.

Last data received: 26/07/2026, 09:23:24

Content source: Microsoft Entra ID, Version: 1.0.0

Author: Microsoft, Supported by: Microsoft Corporation | Email

Related content: 0 Workbooks, 2 Queries, 73 Analytics rules templates

Data received: Go to log analytics

Prerequisites

To integrate with Microsoft Entra ID make sure you have:

- ✓ **Workspace:** read and write permissions.
- ✓ **Diagnostic Settings:** read and write permissions to Microsoft Entra ID diagnostic settings.
- ✓ **Tenant Permissions:** 'Global Administrator' or 'Security Administrator' on the workspace's tenant.

Table management

Manage the table tiers - analytics and lake and table retention. Select the table to manage the table tier and configure its retention settings.

Table name	Tier	Table type	Analytics reten...	Total retention	Workspace
☐ AuditLogs	Analytics	Sentinel	30 days	30 days	law-security-lab
☐ SigninLogs	Analytics	Sentinel	30 days	30 days	law-security-lab

Configuration

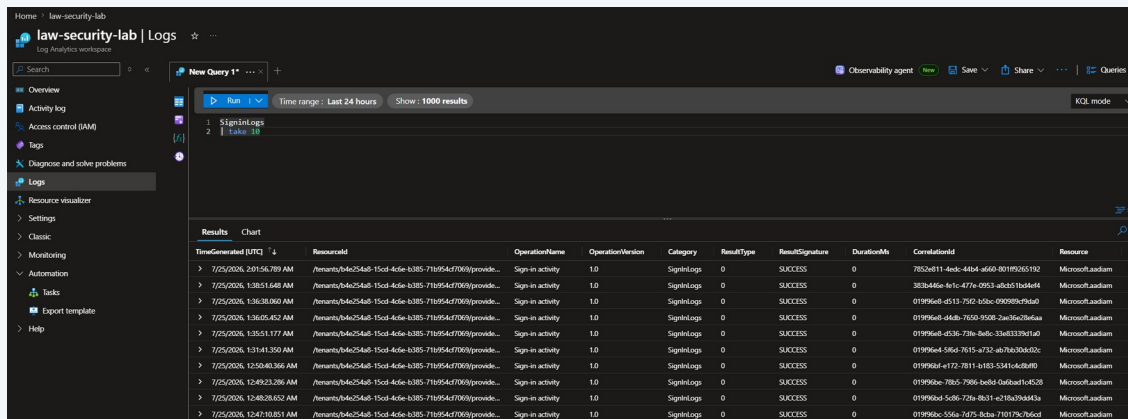
Connect Microsoft Entra ID logs to Microsoft Sentinel

Select Microsoft Entra ID log types:

- ☒ Sign-In Logs
 - ☐ In order to export Sign-in data, your organization needs Microsoft Entra ID P1 or P2 license. If you don't have a P1 or P2, start a free trial.
- ☒ Audit Logs
 - ☐ Non-Interactive User Sign-In Log
 - ☐ Service Principal Sign-In Logs
 - ☐ Advanced Audit Log Settings

Confirming Data Ingestion

- Ran a KQL query (SignInLogs | take 10) before building any detection logic
- Verified logs were flowing end-to-end from Entra ID into the workspace
- Build on confirmed data, not assumptions



The screenshot shows the Microsoft Sentinel Log Analytics workspace interface. The left sidebar contains navigation options: Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Logs (selected), Resource visualizer, Settings, Classic, Monitoring, Automation, Links, and Help. The main pane displays a KQL query named 'New Query 1*' with the following query:

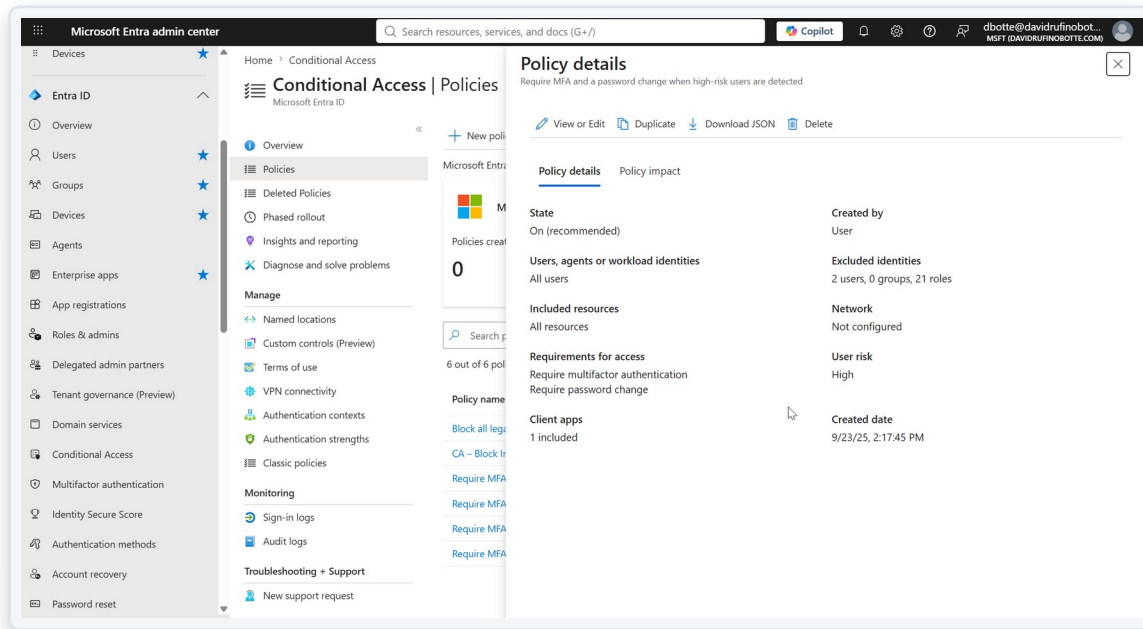
```
1 SignInLogs
2 | take 10
```

The query results are displayed in a table with the following columns: Results, Chart, TimeGenerated [UTC], ResultId, OperationName, OperationVersion, Category, ResultType, ResultSignature, DurationMs, CorrelationId, and Resource. The table shows 10 rows of data, all with a ResultType of 'SignInLog' and a ResultSignature of '0'. The Resource column indicates that the logs are from 'Microsoft.Laasem'.

Results	Chart	TimeGenerated [UTC]	ResultId	OperationName	OperationVersion	Category	ResultType	ResultSignature	DurationMs	CorrelationId	Resource
>		7/25/2024, 2:01:55.789 AM	7f0a0b0c-150d-4d6e-b385-71095d670000	Sign-in activity	1.0	SignInLog	0	0	0	7832d111-4e4d-4d6e-b385-71095d670000	Microsoft.Laasem
>		7/25/2024, 1:28:51.648 AM	7f0a0b0c-150d-4d6e-b385-71095d670000	Sign-in activity	1.0	SignInLog	0	0	0	383d4d6e-b385-477e-09d3-4d6e-b385-71095d670000	Microsoft.Laasem
>		7/25/2024, 1:16:38.900 AM	7f0a0b0c-150d-4d6e-b385-71095d670000	Sign-in activity	1.0	SignInLog	0	0	0	01999d6e-b385-71095d670000	Microsoft.Laasem
>		7/25/2024, 1:16:05.402 AM	7f0a0b0c-150d-4d6e-b385-71095d670000	Sign-in activity	1.0	SignInLog	0	0	0	01999d6e-b385-71095d670000	Microsoft.Laasem
>		7/25/2024, 1:16:51.177 AM	7f0a0b0c-150d-4d6e-b385-71095d670000	Sign-in activity	1.0	SignInLog	0	0	0	01999d6e-b385-71095d670000	Microsoft.Laasem
>		7/25/2024, 1:14:11.350 AM	7f0a0b0c-150d-4d6e-b385-71095d670000	Sign-in activity	1.0	SignInLog	0	0	0	01999d6e-b385-71095d670000	Microsoft.Laasem
>		7/25/2024, 12:59:40.365 AM	7f0a0b0c-150d-4d6e-b385-71095d670000	Sign-in activity	1.0	SignInLog	0	0	0	01999d6e-b385-71095d670000	Microsoft.Laasem
>		7/25/2024, 12:49:23.286 AM	7f0a0b0c-150d-4d6e-b385-71095d670000	Sign-in activity	1.0	SignInLog	0	0	0	01999d6e-b385-71095d670000	Microsoft.Laasem
>		7/25/2024, 12:40:28.052 AM	7f0a0b0c-150d-4d6e-b385-71095d670000	Sign-in activity	1.0	SignInLog	0	0	0	01999d6e-b385-71095d670000	Microsoft.Laasem
>		7/25/2024, 12:41:10.851 AM	7f0a0b0c-150d-4d6e-b385-71095d670000	Sign-in activity	1.0	SignInLog	0	0	0	01999d6e-b385-71095d670000	Microsoft.Laasem

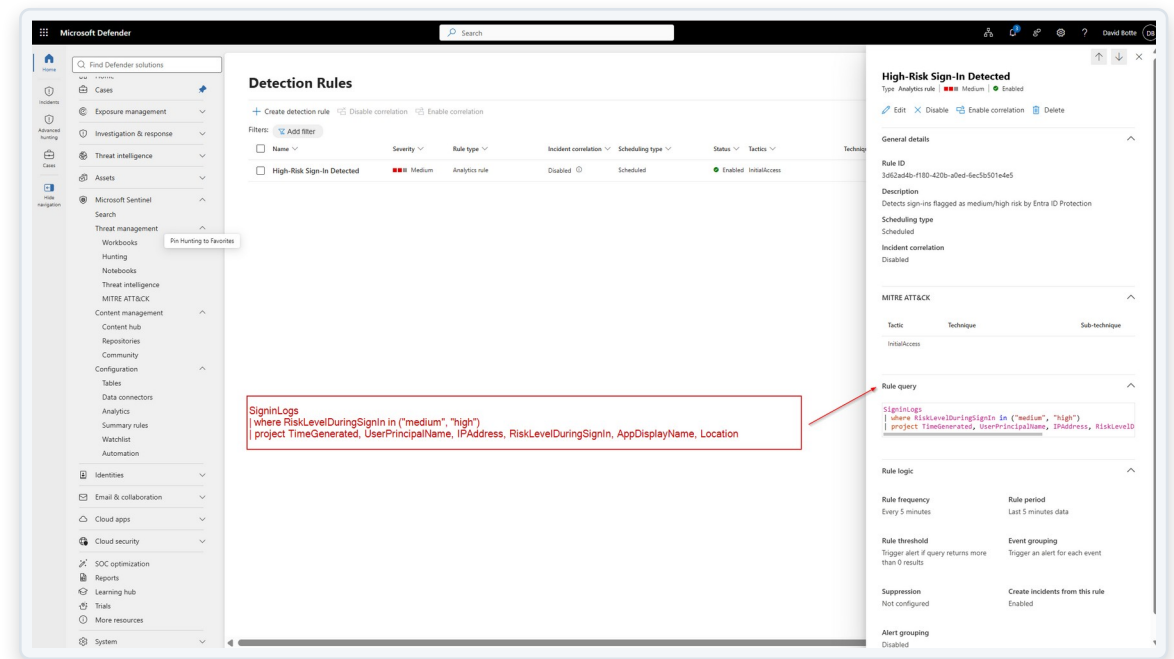
Reusing Existing Controls

- Audited existing Conditional Access policies before creating a new one
- Found and reused a policy already enforcing MFA on risky sign-ins
- Avoids policy sprawl — a real security engineering judgment call



Building the Analytics Rule

- Scheduled KQL rule 'High-Risk Sign-In Detected', running every 5 minutes
- Filters on RiskLevelDuringSignIn (medium / high) from Entra ID Protection
- Mapped to MITRE ATT&CK — Initial Access



Detections at Scale

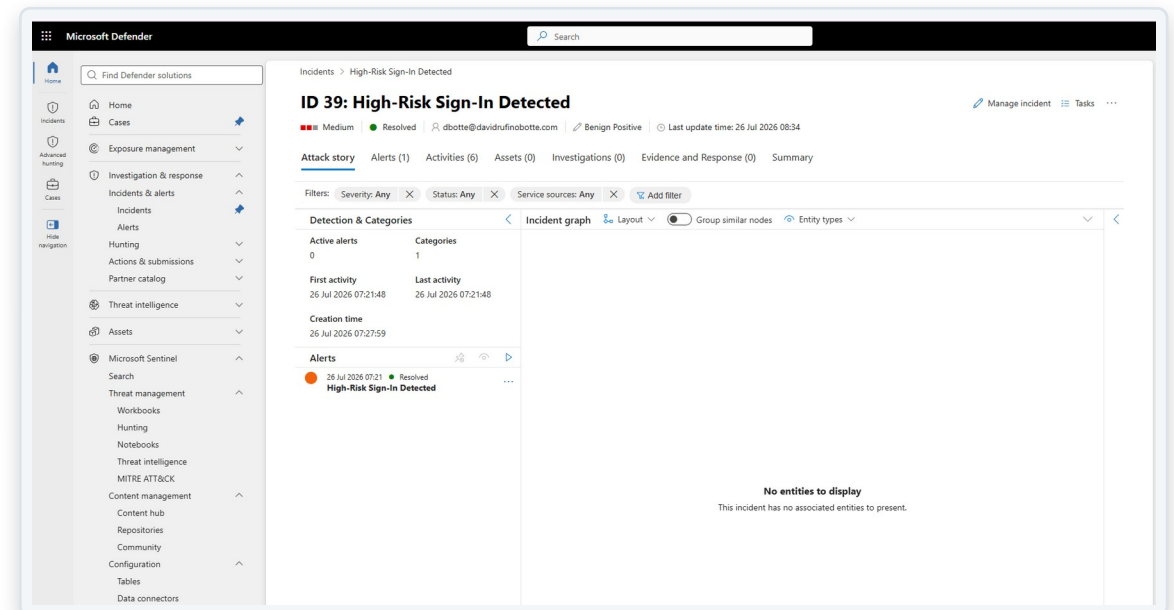
- The rule generated 39 incidents from real sign-in activity
- Confirms the detection logic works against live data, not just theory
- Each incident automatically linked back to its source alert

The screenshot displays the Microsoft Defender 'Incidents' page. The left sidebar shows navigation options like Home, Cases, Exposure management, Investigation & response, Hunting, Threat intelligence, Assets, Microsoft Sentinel, and Cloud apps. The main area is titled 'Incidents' and shows a list of 39 incidents. A filter bar at the top of the list indicates 'Status: Resolved' and 'Alert severity: High, Medium, Low'. The table columns include Incident name, Incident ID, Priority, Severity, Investigation state, Categories, Impacted assets, Active alerts, and Service sources. All incidents are 'High-Risk Sign-In Detected' with a severity of 'Medium' and an investigation state of 'Initial access'.

Incident name	Incident ID	Priority	Severity	Investigation state	Categories	Impacted assets	Active alerts	Service sources
> High-Risk Sign-In Detected	39	2	Medium	Initial access			0/1	Microsoft Sentinel
> High-Risk Sign-In Detected	38	2	Medium	Initial access			0/1	Microsoft Sentinel
> High-Risk Sign-In Detected	37	2	Medium	Initial access			0/1	Microsoft Sentinel
> High-Risk Sign-In Detected	35	2	Medium	Initial access			0/1	Microsoft Sentinel
> High-Risk Sign-In Detected	34	2	Medium	Initial access			0/1	Microsoft Sentinel
> High-Risk Sign-In Detected	36	2	Medium	Initial access			0/1	Microsoft Sentinel
> High-Risk Sign-In Detected	33	2	Medium	Initial access			0/1	Microsoft Sentinel
> High-Risk Sign-In Detected	31	2	Medium	Initial access			0/1	Microsoft Sentinel
> High-Risk Sign-In Detected	32	2	Medium	Initial access			0/1	Microsoft Sentinel
> High-Risk Sign-In Detected	30	2	Medium	Initial access			0/1	Microsoft Sentinel
> High-Risk Sign-In Detected	29	2	Medium	Initial access			0/1	Microsoft Sentinel
> High-Risk Sign-In Detected	28	2	Medium	Initial access			0/1	Microsoft Sentinel
> High-Risk Sign-In Detected	27	2	Medium	Initial access			0/1	Microsoft Sentinel
> High-Risk Sign-In Detected	26	2	Medium	Initial access			0/1	Microsoft Sentinel
> High-Risk Sign-In Detected	25	2	Medium	Initial access			0/1	Microsoft Sentinel

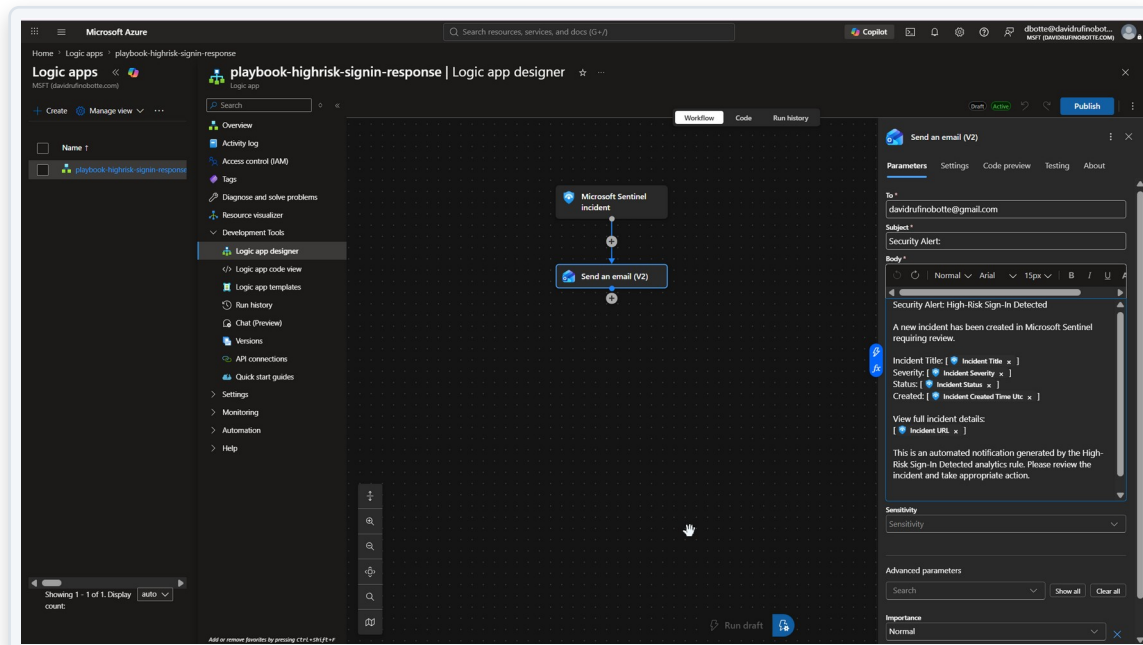
Incident Review & Closure

- Reviewed and closed incidents individually with correct classification
- Classified as Benign Positive — the detection was correct; the activity was a controlled test
- Standard SOC discipline, not blanket dismissal



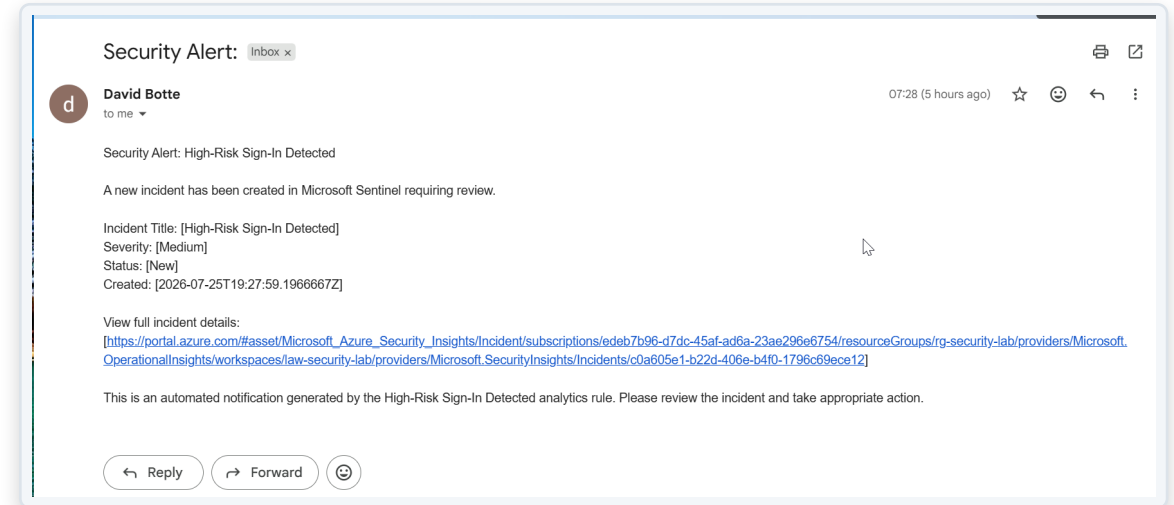
Automated Response Playbook

- Logic App triggered directly by 'Microsoft Sentinel incident'
- Sends a structured alert with incident title, severity, status and link
- Consumption plan used deliberately, to avoid fixed hosting cost



Closing the Loop

- Alert delivered by email within minutes of incident creation
- Full pipeline proven: sign-in log → detection → incident → automated notification
- No manual step between detection and alert delivery



WHAT THIS DEMONSTRATES

Skills applied in this lab

- **Identity Security**

Entra ID, Conditional Access, risk-based sign-in policy

- **Security Automation**

Logic Apps / SOAR-style automated incident response

- **Troubleshooting**

Diagnosed and resolved real platform issues (missing connector, permissions, cross-tenant setup)

- **SIEM Engineering**

Microsoft Sentinel, KQL detection rules, MITRE ATT&CK mapping

- **SOC Discipline**

Incident triage, classification, and closure — not just detection

- **Cost & Governance Awareness**

Reused existing controls; chose consumption pricing to avoid unnecessary cost